



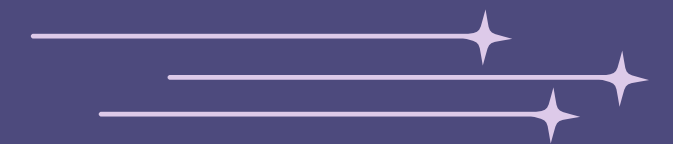
CS496 -Zero Knowledge proof: From theory to application

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Introduction

This project studies Zero-Knowledge Proofs (ZKPs) starting from their formal foundations and exploring their vastness by Oded's Paper, connecting formal proofs in first order logic in Model theory in mathematics to coloring of humongous graph .Then Proceeds to implement interactive ZKPs for Sudoku and Graph 3-Coloring using SHA-256 commitments and C++ Asio networking.



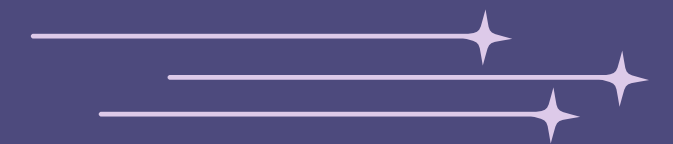


Theory

Definition of ZKP :

Zero-knowledge proofs are proofs that yield nothing beyond the validity of the assertion. That is, a verifier obtaining such a proof only gains conviction in the validity of the assertion.

But isn't it counter intuitive😞! What one mean by this?



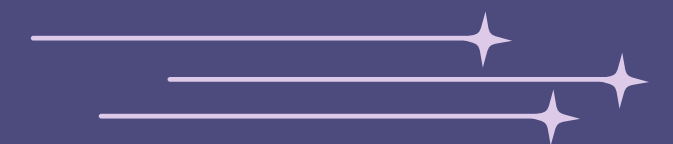
What are proofs?

Mathematical proof of theorem is sequence of arguments such that it convinces one reads it about the validity of the statement.

ZKPs?

In this proof we share knowledge to reader or verifier. In Zero Knowledge Proof, our task is to give conviction to reader or verifier about the statement but we do not want to share any knowledge to them.

Let see, how we can do it



Graph Coloring Problem

Given a graph , we have to provide its coloring using 3 colors such that no two same color nodes are adjacent to each other.

Now, suppose Alice wants to prove Bob that he knows the coloring of given graph, but she does not trust Bob to share him her knowledge.

Alice proceeds with Strategy:

Bob randomly asks to check for any random pair of adjacent nodes.

Alice reveals.

Bob verifies.

She then randomly permutes coloring out of $3!$ permutations. Repeat..



Statistical analysis

Suppose there are m edges in graph,
if Alice do not know how to color, then there exist edge she is making
mistake.

$P(\text{Bobs can not verify that Alice is cheating}) = \left[\frac{(m-1)}{m} \right]^n$
where n is number of trials. As n goes to ∞ . implies probability tends to
0 i.e. Alice can not cheat.



How to check proof is not yielding any knowledge to verifier?

If that proof can also be generated by someone who does not anything or say any simulator, then we are assured that it is not leaking any knowledge.

e.g. In previous proof, reader/verifier only knows that two adjacent colors are different.



Which statements can have Zero Knowledge proof?

By Paper “Proof that yield nothing but their Validity.... ” published by Oded Goldreich, Silvio Micali and Avi Wigderson– For roughly every theorem one can have Zero Knowledge Proof.

Can we do it for proof of Fermat's last theorem by Wiles? Yes, it's the beauty of Graph coloring Problem.



So, the way mathematical proofs are written they are not completely formal e.g. writing “see this proof for this”,etc.. Writing this formally means we write it in language of 1st order logic.

This would make Wiley’s Paper from 300 pages to 1 million pages

1.	$\forall x(A(x) \rightarrow B) \vdash \forall x(A(x) \rightarrow B)$	Assumption
2.	$\forall x(A(x) \rightarrow B) \vdash \forall x(\neg B \rightarrow \neg A(x))$	Exercise
3.	$\forall x(A(x) \rightarrow B) \vdash \neg B \rightarrow \forall x \neg A(x)$	Axiom 5
4.	$\forall x(A(x) \rightarrow B) \vdash \neg \forall x \neg A(x) \rightarrow B$	PC 3
5.	$\forall x(A(x) \rightarrow B) \vdash \exists x A(x) \rightarrow B$	Definition of $\exists$
6.	$\vdash \forall x(A(x) \rightarrow B) \rightarrow (\exists x A(x) \rightarrow B)$	Deduction
7.	$\exists x A(x) \rightarrow B \vdash \exists x A(x) \rightarrow B$	Assumption
8.	$\exists x A(x) \rightarrow B \vdash \neg \forall x \neg A(x) \rightarrow B$	Definition of $\exists$
9.	$\exists x A(x) \rightarrow B \vdash \neg B \rightarrow \forall x \neg A(x)$	PC 8
10.	$\exists x A(x) \rightarrow B \vdash \forall x(\neg B \rightarrow \neg A(x))$	Theorem 8.19

We then convert these statements into graph coloring.

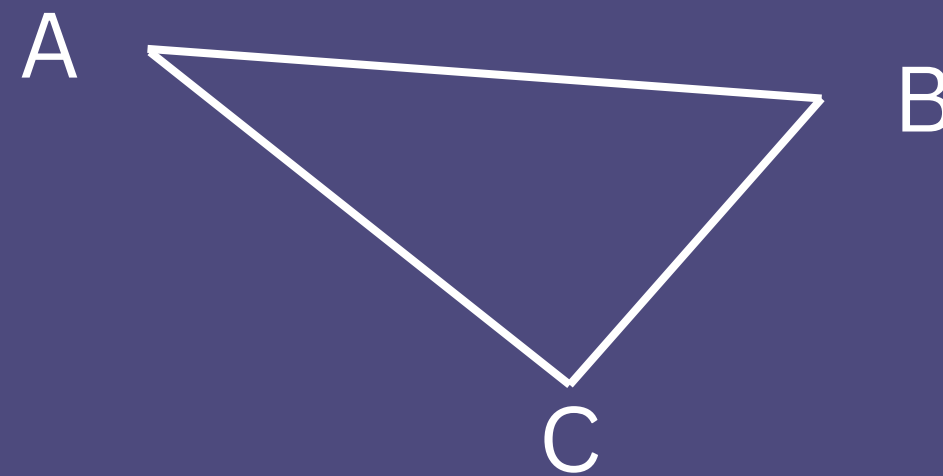
1st order logic language



For example consider this,

$$\neg(A = B) \wedge \neg(A = C) \wedge \neg(B = C)$$

Proving it correct is equivalent to say color this graph



NP completeness is Fundamental point

It allows us when I want to prove a theorem like “ZKP for every statement..” and also in every other situation. Also In complex theory, having a particular problem with nice properties that captures every other problem is very convenient.



Problem is NP complete when:

- A non deterministic turing machine can solve it in polynomial time.
- A deterministic turing machine can solve it in large time and can verify its solution in polynomial time.
- It can be used to simulate any other problem with similar solvability.



Applications



Software Implementation of ZKP for Graph Coloring

1. Commitment Scheme: Prover use cryptographic hash function to commit to its coloring and send all hash values to verifier.
2. Challenge: Verifier challenges by random asking colors for nodes connecting any edge.
3. Reveal: Prover reveals colour and random string he/she used in hashing.
4. Verify: Verifier verifies those values with its hash value.



Networking of Programs of Prover and verifier

I used asio.hpp library and its socket in both programs through which they can communicate with each other.

While sharing vector of strings we have to share one by one otherwise we can not understand bits we are getting from side.

For sharing integer numbers which are of 4 byte , I used function htonl and ntohl ,for converting it from host byte order to network byte order and vice versa.



Random Number generating and random shuffle

Used library <random> to use random_device, mt19937 and dis function to generate random numbers and converting them into strings.

Used shuffle function for randomly shuffling permutations.



ZKP for Sudoku problem

Aim: Alice wants to give ZKP that she knows how to solve given sudoku as she does not trust Alice.

Our Strategy:

1. Sudoku has constraint that no same number should be in same column, same row and same block.
2. So, instead of showing whole grid, Alice can show some numbers in either same row, same column or same block to Bob and then random permutes them for next round.
3. By Similar Statistical analysis , it can be proven that no one of them can cheat.



Any Questions?

Thank you